



Alberta Public Health Disease Management Guidelines

Chikungunya Virus (CHIKV)



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Public Health Division

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without [clinical illness](#)^(A) with:

- Detection of Chikungunya virus (CHIKV) nucleic acid (e.g., by RT-PCR) in serum or plasma, or cerebrospinal fluid (CSF)^(B)

OR

- Viral culture of CHIKV in serum^(C) or CSF

OR

- Detection of CHIKV specific IgM with confirmation by a positive Plaque Reduction Neutralization Test (PRNT) (titer of 1:20 or greater), or demonstration of IgG seroconversion on acute-convalescent paired sera

^(A) Clinical illness is characterized by a febrile illness of variable severity associated with arthritis-like pain in the joints. Other symptoms may include fatigue, headache, nausea, vomiting, muscle pain, or rash. For more information, refer to [clinical presentation section](#)

^(B) In patients with neurological symptoms

^(C) **NOTE:** Serum is preferred to plasma

Reporting Requirements

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the Zone Medical Officer of Health (MOH) (or designate) and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Chikungunya virus (CHIKV) is a single-strand, positive-sense RNA alphavirus belonging to the *Togaviridae* family.^(1,2) There are four known genotypes - Asian; West African; East, Central, and South African; and Indian Ocean Lineage.^(3,4)

Clinical Presentation

Most people infected with CHIKV will develop some symptoms. CHIKV infection starts with a sudden onset of fever (temperature 39°C to 40°C) that lasts several days and up to one week. Illness is often characterized by arthralgia and a maculopapular rash. Joint pain involves multiple joints and commonly occurs in hands and feet but can also occur in joints of the upper extremities, and the ankles and knees. The joint pain can be severe and debilitating. Rash can be pruritic; usually occurs after onset of fever and can be observed on the face, palms, soles, extremities and trunk.

Other symptoms may include headache, nausea, vomiting, anorexia, myalgia, joint swelling and conjunctivitis. Acute symptoms usually resolve within seven to 10 days; however, some individuals may experience a relapse or persistence of symptoms such as joint pain and myalgia in the weeks, months or years after initial infection. Although rare, hemorrhagic manifestations can occur in approximately 5% of cases.^(2,5,6)

Neurological, ophthalmological, cardiac, hemorrhagic and gastrointestinal complications of CHIKV have been reported.^(7,8) Severe cases and fatalities from CHIKV infection are rare, but have been reported especially during outbreaks and are most often seen in older adults (≥65 years), individuals with underlying comorbidities and newborns infected around the time of birth.^(2,8–10)

Diagnosis

A preliminary diagnosis is usually based on exposure (e.g., history of travel to an endemic area) and clinical symptoms. Since CHIKV, zika and dengue viruses are in co-circulation in some areas, and there are similarities in symptoms with infection of these viruses, laboratory testing is important.^(8,11)

Nucleic acid amplification or viral culture performed in the first week after onset of illness can be used to diagnose CHIKV infection.^(2,7,8) Laboratory confirmation can also be accomplished by detecting virus-specific IgM and neutralizing antibodies in serum that develop four to five days after symptom onset. Individuals who initially test sero-negative in the early days of infection should have convalescent-phase samples taken. Levels of IgM antibodies are high 3 to 5 weeks after symptom onset and can last for months. There have been reports of serological cross-reactions between other arboviruses such as dengue and zika, therefore plaque-reduction neutralization tests (PRNT) is required to measure virus specific antibodies and differentiate between cross-reacting antibodies.⁽¹²⁾

Treatment

No specific antiviral treatment has been identified for CHIKV. Analgesics and antipyretics are useful to manage symptoms such as joint pain and fever. Symptomatic individuals are advised to get plenty of rest and to drink fluids.^(7,8)

Reservoir

During epidemic periods, humans are the main reservoir for CHIKV. However, rodents, monkeys, birds and bats serve as the natural reservoir outside these periods.⁽¹³⁾

Transmission

Infection is transmitted to humans through the bite of an infected *Aedes aegypti* or *Aedes albopictus* mosquito.⁽⁹⁾ The mosquito becomes infected after feeding on a source infected with the virus and then spreads the infection in subsequent blood meals. Although rare, transmission in utero has been reported.⁽¹⁴⁾ Perinatal transmission can occur during delivery in women who developed illness within the final week prior to delivery. Blood-borne transmission of CHIKV has also been reported in a health care provider who sustained a needle stick injury after drawing blood from an infected individual and laboratory technicians handling blood specimens infected with CHIKV.^(2,12,14)

Incubation Period

The incubation period is typically 3–7 days (range, 1–12 days).⁽¹¹⁾

Period of Communicability

Individuals infected with CHIKV are viremic from two days before symptom onset and in the first week after illness onset.^(2,15,16)

Host Susceptibility

Susceptibility is universal. Older adults, individuals with underlying comorbidities and newborns infected around the time of birth are at increased risk for severe illness and death. Individuals who have been infected and recovered from CHIKV experience life-long immunity against future infection.^(7,9)

Incidence

CHIKV was first isolated and identified in Tanzania during an outbreak in 1952-1953 and since then cases and outbreaks have been identified globally. It is endemic in Africa, Southeast Asia, the Indian subcontinent, Pacific Region and in the (sub) tropical regions of the Americas.^(7,8) Up to date case counts by region can be found [here](#). The risk of CHIKV infection in Canada and Alberta is low and reported cases are related to travel to areas where CHIKV is circulating and/or endemic. For current cases reported in Alberta, refer to the [Interactive Health Data Application \(IHDA\)](#).

Public Health Management

Key Investigation

- Confirm the diagnosis and rule out other arboviral infections such as dengue and zika.
- Rule out other infections or conditions that may cause similar symptoms, e.g., malaria, group A streptococcus, leptospirosis, measles, enterovirus, rubella, parvovirus, adenovirus, rheumatologic conditions, alphavirus infections and post-infections arthritis, including coinfection in returning travelers.⁽²⁾
- Determine exposure history, taking into consideration the incubation period, reservoir, and mode of transmission, including:
 - recent travel to an endemic area, or
 - recent emigration from an endemic area.

Management of a Case

- There is no specific antiviral therapy.
- Management is mainly treatment of symptoms with analgesics, antipyretics, drinking fluids and rest. Consultation with a specialist may be considered if symptoms such as arthritis last more than three months.
- To prevent further transmission, cases should avoid further mosquito bites during the first week of illness.

Management of Contacts

- There is no follow-up of contacts.
- It may be prudent to identify other symptomatic individuals who may have been exposed to the same source as the case (i.e., co-travelers) to identify undiagnosed cases.

Preventive Measures

- No vaccine or medication is available to prevent CHIKV infection or disease.
- Recommendations for persons travelling to endemic areas include:
 - Consult a health care provider or visit a travel health clinic, preferably six weeks before travel.
 - Use personal protective measures to reduce mosquito exposure, for example:
 - stay in well-screened or completely enclosed, air-conditioned areas whenever possible.
 - sleep under a bed net, preferably one treated with insecticide.
 - wear light-coloured clothing and cover up as much as possible with long pants, long-sleeved loose fitting shirts, shoes/boots (not sandals) and a hat.
 - apply mosquito/insect repellent (containing DEET or Icaridin/picaridin) on exposed skin and on clothing, 15 minutes after applying sunscreen.
 - reduce mosquito breeding sites in outdoor/indoor areas by draining or discarding sources of standing water (e.g., open-air containers such as flower pots, bottles, used tires and rock pools).
- Individuals with suspected CHIKV infection should avoid further mosquito exposure during the first week of illness to reduce the risk of forward transmission.^(8,17)
- Individuals who are at increased risk for severe illness, such as older adults with underlying medical conditions, should consider not traveling to areas with ongoing Chikungunya transmission or outbreaks.^(7,8,11)

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	Updated Template
December 2022	Case Definition	Suspect case definition removed as not reportable
	General	- Entire guideline updated to include information on sections related to Epidemiology and Public Health Management - Updated references

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